SYNOPSIS Presentation

on

FaceTally: Facial Attendance System

by

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PROJECT OVERVIEW

What is this Project?

- A camera-based AI system that automates student attendance using facial recognition.
- Built using Python, FaceNet deep learning, SVM classification, and MySQL database.
- Replaces manual roll calls with real-time, contactless, automated attendance logging.

Problem it Solves

- Manual attendance is time-consuming, error-prone, and susceptible to proxy fraud.
- Biometric and RFID systems are contact-based or easily misused.
- No real-time tracking or automated record management in traditional methods.

Significance & Impact

- Eliminates proxy attendance through biometric-level face verification.
- Saves faculty time and improves institutional data accuracy.
- Supports SDG 4 — Quality Education through smart technology adoption.

PROJECT OBJECTIVES

Automate Attendance

- Develop a real-time face recognition system to mark attendance automatically.
- Eliminate the need for manual roll calls by faculty.

Eliminate Proxy

- Prevent proxy attendance using facial biometrics that cannot be shared.
- Block duplicate attendance entries for the same student.

Accurate Record Management

- Store and manage student records and logs in a MySQL database.
- Enable easy retrieval, search, and filtering of attendance data.

Measurable Success

- Achieve face recognition accuracy ≥ 95% under normal lighting.
- Reduce attendance marking time from ~5 minutes to under 30 seconds.

Success Metrics: Recognition Accuracy ≥ 95% | Attendance Logged in < 2 sec | Zero Proxy Incidents | 100% Duplicate Prevention

PROJECT SCOPE

What's Included

- Student registration with webcam-captured face photos
- Real-time face detection via FaceNet + SVM
- Automatic attendance logging with date and time
- Admin dashboard with 6 functional modules
- MySQL database integration for reports

What's Excluded

- Mobile or web-based interface (Desktop Only)
- Multi-camera simultaneous support
- Cloud hosting or remote database access
- Anti-spoofing / liveness detection (Future Goal)

Key Modules

Module	Function
Admin Login	Secure credential-based access
Student Mgmt	Register, update, delete students
Extraction	Generate FaceNet embeddings
Training	Train SVM on embeddings
Attendance	Real-time webcam recognition
Reports	View, search, filter logs

METHODOLOGY

Development Approach

Methodology: Agile Model

Project developed in iterative phases — each module built, tested, and refined independently. Continuous testing ensures bug-free integration.

System Workflow

Register Student info → **Extract** Features → **Train** SVM Model → **Detect** Recognize → **Log** Attendance

Tools & Technologies

Category	Technology
Language	Python 3.10
GUI	Tkinter
Vision	OpenCV
DL	TensorFlow / Keras
Embedding	FaceNet
Classifier	SVM (scikit-learn)
Database	MySQL (XAMPP)

HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i5 / i7 (Min 2.0 GHz)
- RAM: Minimum 8 GB (16 GB Recommended)
- Storage: Minimum 10 GB free space
- Webcam: USB or Built-in (720p Minimum)
- GPU: Optional NVIDIA (for faster inference)
- Display: 1366 × 768 Resolution

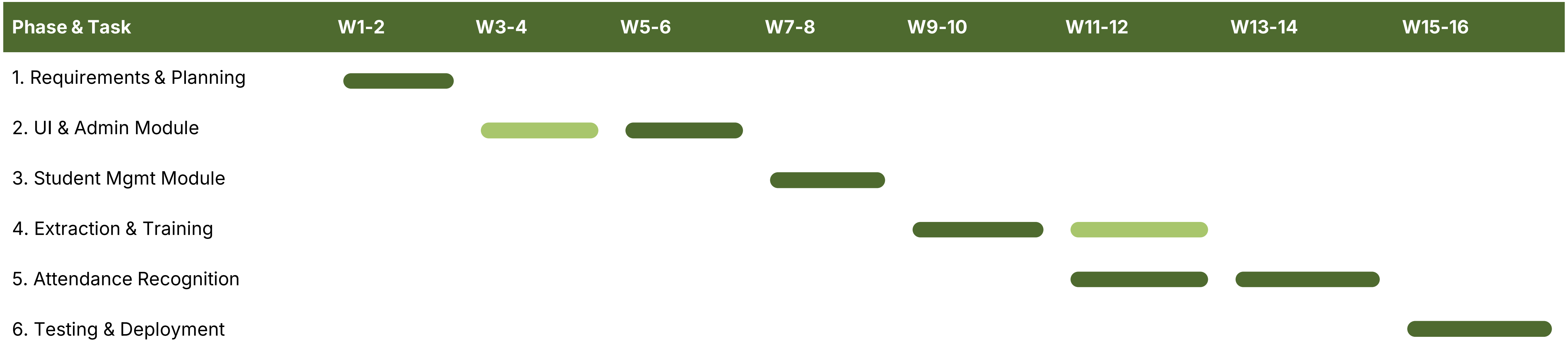
Software & Purpose

Software	Purpose
Python 3.10	Core Language
OpenCV	Face Detection
TensorFlow	DL Framework
Scikit-learn	SVM Classifier
Tkinter	Desktop UI
MySQL	Database Mgmt

OS Compatibility: Windows 10/11 (Primary) | Linux Ubuntu 20.04+ (Supported) | Virtual Env Recommended

PROJECT TIMELINE

The project follows a 6-phase incremental development plan spanning approximately 16 weeks.



🏁 **Week 4 — Prototype Ready**

🏁 **Week 10 — Core System Complete**

🏁 **Week 16 — Final Delivery Done**

PROJECT OUTCOME

Expected Deliverables & Achievements

Desktop Application

Fully functional Python-Tkinter app with 6 integrated modules, deployed in lab environment.

Impact Summary: The system is projected to reduce attendance marking time by 90%, eliminate proxy fraud entirely, and serve as a scalable model for institutional smart infrastructure.